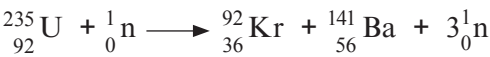


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4. (a) A fission nuclear reaction for uranium is:



The masses of the nuclei are:

mass of ${}^{235}_{92}\text{U}$ = 235.01 u

mass of ${}^{92}_{36}\text{Kr}$ = 91.90 u

mass of ${}^{141}_{56}\text{Ba}$ = 140.89 u

mass of ${}^1_0\text{n}$ = 1.01 u

1 u = 931 MeV

Calculate the energy released in this nuclear reaction.

[2]

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(b) Explain carefully the relevance of the **binding energy per nucleon** curve to nuclear fission and fusion. [6 QER]

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